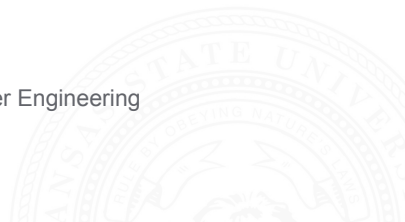


# Three-Phase Transformers I

Windings, configurations, and phase shift

**Buxin She**

Mike Wieggers Department of Electrical and Computer Engineering



# Lecture outline

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- 1 Winding count, phase count, and physical construction.
- 2 Coil connections for four three-phase banks.
- 3 Terminal relations and phase shift, followed by a numerical bank example.

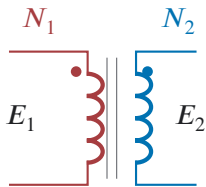
**Follow each connection from its coils to its terminal phasors.**

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## PART 1

# Windings and phases

# Winding and magnetic coupling



two isolated winding circuits

Both coils share core flux  $\Phi$ .

$$\frac{E_1}{N_1} = \frac{E_2}{N_2}$$

## Winding and voltage symbols

A winding is a coil of insulated conductor.

$N_1, N_2$ : turns in windings 1 and 2.

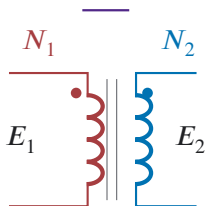
$E_1, E_2$ : induced RMS voltage phasors.

$V_{xy}$ : RMS terminal-voltage phasor;  $x$  relative to  $y$ .

**For an ideal winding,  $E = V$  at its terminals with the same polarity reference.**

# Two-winding and three-winding

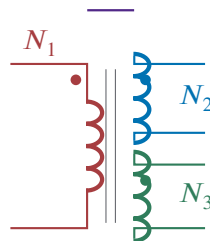
## Two-winding



two isolated winding circuits

$$E_1/N_1 = E_2/N_2$$

## Three-winding



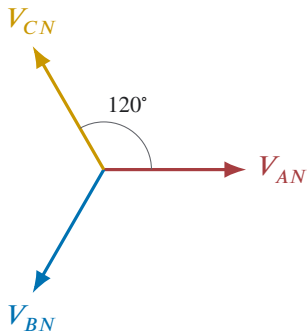
three isolated winding circuits

$$E_1/N_1 = E_2/N_2 = E_3/N_3$$

**Three-winding adds a third voltage circuit coupled to the same flux.**

Source: Glover et al., sixth edition (2017), Sec. 3.6, Fig. 3.20

# Phase and phase systems



$A, B, C$ : supply-phase terminals.  
 $N$ : common phase-voltage reference.

## Voltage notation

$v(t)$ : instantaneous voltage.

$V$ : RMS magnitude;  $\phi$ : phase angle.

$f$ : frequency;  $\omega = 2\pi f$ : angular frequency.

$$v(t) = \sqrt{2}V \cos(\omega t + \phi)$$

Balanced positive sequence:

$$V_{AN} = V \angle 0^\circ$$

$$V_{BN} = V \angle (-120^\circ)$$

$$V_{CN} = V \angle 120^\circ$$

Use  $E$  for induced winding voltage and  $V_{xy}$  for voltage between terminals.

# Winding count and phase count

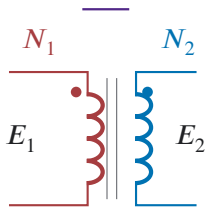
|               | Single-phase    | Three-phase                   |
|---------------|-----------------|-------------------------------|
| Two-winding   | 2 coil circuits | 2 winding sets; 6 phase coils |
| Three-winding | 3 coil circuits | 3 winding sets; 9 phase coils |

For the conventional constructions shown here, each three-phase winding set contains one coil circuit per phase.

**A three-phase two-winding transformer has two winding sets, each with three phase coils.**

# Single-phase and three-phase

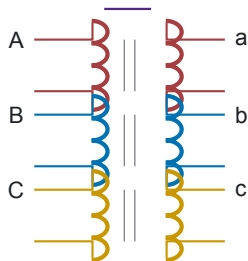
## Single-phase



two isolated winding circuits

One alternating voltage system.

## Three-phase

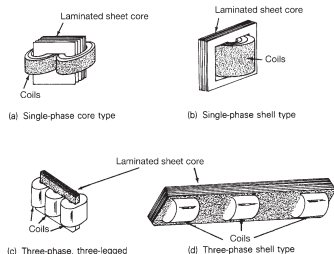


$A/a$ ,  $B/b$ ,  $C/c$ : matching phase-coil pairs.

Supply phases are separated by  $120^\circ$ .

**Three-phase operation requires three corresponding phase winding pairs.**

# Bank and common core



## Two physical implementations

A bank uses three separate single-phase transformers.

A three-phase unit places the winding sets on a common three-phase core.

Both can realize the same terminal connection.

**Core construction and external Y/ $\Delta$  connection are separate choices.**

Source: Glover et al., sixth edition (2017), Fig. 3.15

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## PART 2

# Connections and phase shift

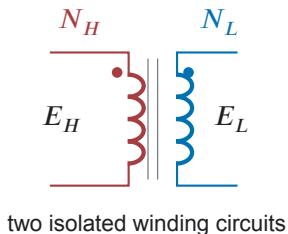
# Terminal and phase conventions

- 1 H and L now identify high- and low-voltage sides; their coil quantities use H/L subscripts.
- 2 H phases are  $A, B, C$ ; L phases are  $a, b, c$ .
- 3 Terminal marks:  $H_1, H_2, H_3$  on H;  $X_1, X_2, X_3$  on L.
- 4 Positive sequence:  $A \rightarrow B \rightarrow C$ ; RMS phasors.
- 5 Compare the corresponding line voltages  $V_{AB,H}$  and  $V_{ab,L}$ .

$$a = \frac{N_H}{N_L} = \frac{|E_H|}{|E_L|}, \quad \delta_{LH} = \angle V_{ab,L} - \angle V_{AB,H}$$

**Winding turns ratio  $a$  and signed terminal displacement  $\delta$  are different quantities.**

# Dots and matching coils



Measure each winding voltage from its dotted to its undotted end.

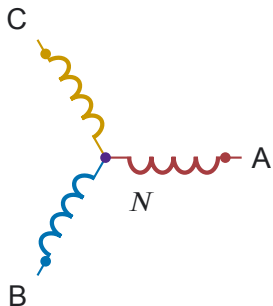
$$E_H = aE_L$$

Same-color coils are magnetically paired.

Wire-junction dots and winding-polarity dots have different roles.

**A dot transfers coil polarity; the external terminal joining determines the bank phase shift.**

# Y: line and coil quantities



## Balanced positive sequence

$V_{AN}$ : voltage across coil  $AN$ .

$I_A = I_{AN}$ : line current equals coil current.

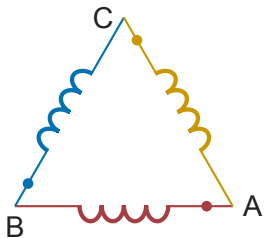
$$V_{AB} = V_{AN} - V_{BN}$$

$$V_{AB} = \sqrt{3}e^{j30^\circ} V_{AN}$$

A star point exists; neutral access requires a brought-out lead.

Line voltage is  $\sqrt{3}$  times phase-coil voltage and leads it by  $30^\circ$ .

# $\Delta$ : line and coil quantities



## Balanced positive sequence

Coil  $AB$  sees line voltage  $V_{AB}$ .

$I_A$  enters A;  $I_{AB} : A \rightarrow B$ ;

$I_{CA} : C \rightarrow A$ .

$$I_A = I_{AB} - I_{CA}$$

$$I_A = \sqrt{3}e^{-j30^\circ} I_{AB}$$

The closed delta has no neutral terminal.

Line current is  $\sqrt{3}$  times coil current and lags it by  $30^\circ$  for the stated references.

# From coils to bank terminals

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## Paired winding voltages

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$$a = \frac{N_H}{N_L} = \frac{|E_H|}{|E_L|}$$

$a$  compares the magnetically paired coils.

## Bank terminal voltages

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$$\frac{|V_{LL,H}|}{|V_{LL,L}|}$$

$LL$ : line-to-line. The terminal ratio compares H/L line-voltage magnitudes.

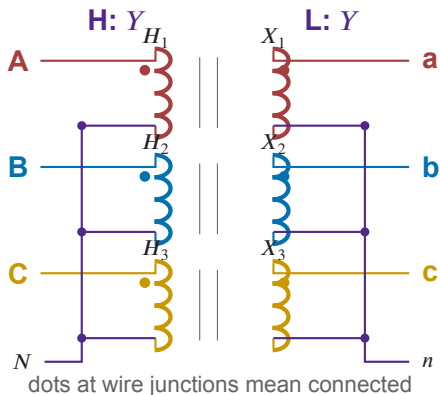
## Use one procedure for every connection

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Trace the paired coil endpoints. Apply the Y/ $\Delta$  line-to-coil relations on each side. Then compare corresponding line-voltage magnitudes and angles.

**First establish the coil joining; then derive what the bank terminals see.**

# Y–Y: physical connection



## Trace the coil endpoints

The H coil ends join at  $N$ ; the L coil ends join at  $n$ .

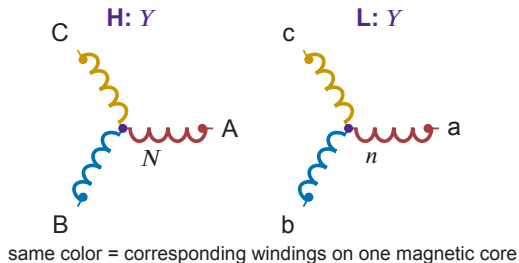
Matching pairs:  $AN/an$ ,  $BN/bn$ ,  $CN/cn$ .

Terminal marks:  $H_1, H_2, H_3$  on H;  
 $X_1, X_2, X_3$  on L.

$X$  is the low-side terminal prefix.

**Both winding sets have a star point.**

# Y–Y: terminal relations



Apply the Y relations on both sides

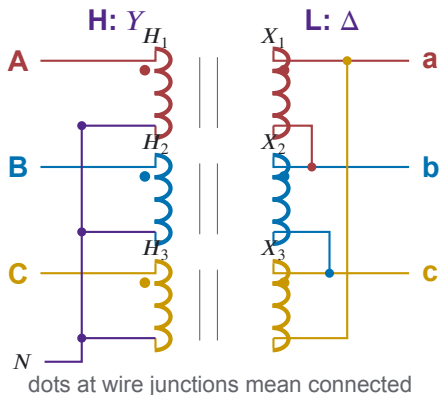
$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{\sqrt{3}|E_H|}{\sqrt{3}|E_L|} = a$$

$$\frac{|I_{L,H}|}{|I_{L,L}|} = \frac{1}{a}$$

$$V_{ab,L} = V_{AB,H}/a, \quad \delta_{LH} = 0^\circ$$

The same Y factors cancel in both magnitude and phase.

# Y- $\Delta$ : physical connection



## Trace the coil endpoints

H coils:  $AN$ ,  $BN$ ,  $CN$ .

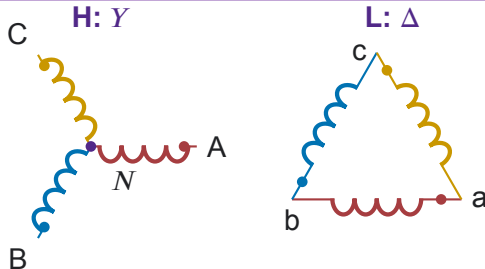
Matching L coils:  $ab$ ,  $bc$ ,  $ca$ .

L delta loop:  $a \rightarrow b \rightarrow c \rightarrow a$  (abc order).

Follow the undotted ends to trace this loop.

**H forms a star; L closes a delta.**

# Y-Δ: terminal relations

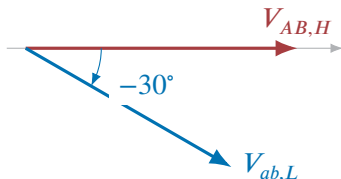


same color = corresponding windings on one magnetic core

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{\sqrt{3}|E_H|}{|E_L|} = \sqrt{3}a, \quad \frac{|I_{L,H}|}{|I_{L,L}|} = \frac{1}{\sqrt{3}a}$$

The H side contributes the Y line-voltage factor; the L side uses coil voltage directly.

# Y-Δ: terminal phase shift



lengths normalized to compare angles

## L delta joining order

Paired with H coils  $AN, BN, CN$ :

$abc: ab, bc, ca \Rightarrow \delta_{LH} = -30^\circ$ .

$acb: ac, ba, cb \Rightarrow \delta_{LH} = +30^\circ$ .

Fixed dots and positive supply sequence.

Use  $\angle V_{AB,H} = 0^\circ$  as reference.

1. H coil  $AN$  (Y):

$$E_H = V_{AN,H} = |E_H| \angle (-30^\circ)$$

2. Transfer to paired L coil  $ab$ :

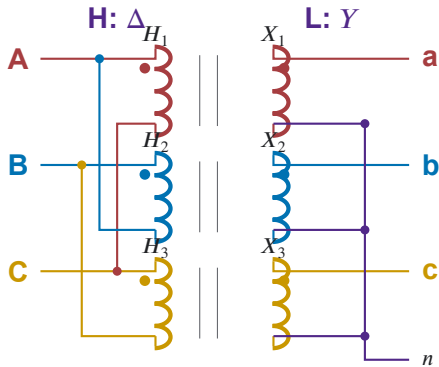
$$E_L = \frac{E_H}{a} = V_{ab,L}$$

3. Form the H line voltage (Y):

$$V_{AB,H} = \sqrt{3} e^{j30^\circ} E_H$$
$$\delta_{LH} = -30^\circ - 0^\circ = -30^\circ$$

**The H line voltage leads its coil voltage by  $30^\circ$ ; the paired L coil is line  $ab$ .**

# $\Delta$ -Y: physical connection



dots at wire junctions mean connected

## Trace the coil endpoints

H coils:  $AC$ ,  $BA$ ,  $CB$ .

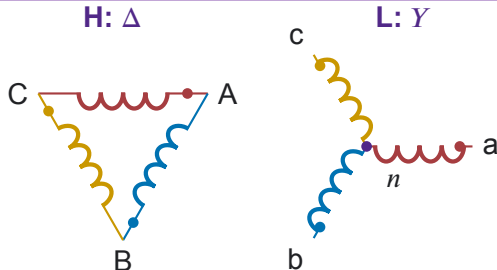
Matching L coils:  $an$ ,  $bn$ ,  $cn$ .

H delta loop:  $A \rightarrow C \rightarrow B \rightarrow A$  (ACB order).

The L coil ends meet at  $n$ .

**H closes a delta; L forms a star.**

# $\Delta$ -Y: terminal relations

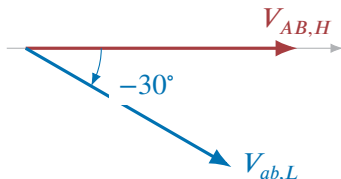


same color = corresponding windings on one magnetic core

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{|E_H|}{\sqrt{3}|E_L|} = \frac{a}{\sqrt{3}}, \quad \frac{|I_{L,H}|}{|I_{L,L}|} = \frac{\sqrt{3}}{a}$$

The L side contributes the Y line-voltage factor in the denominator.

# $\Delta$ -Y: terminal phase shift



lengths normalized to compare angles

## H delta joining order

Paired with L coils  $an, bn, cn$ :

ACB:  $AC, BA, CB \Rightarrow \delta_{LH} = -30^\circ$ .

ABC:  $AB, BC, CA \Rightarrow \delta_{LH} = +30^\circ$ .

Fixed dots and positive supply sequence.

Use  $\angle V_{AB,H} = 0^\circ$  as reference.

1. H coil  $AC$  ( $\Delta$ ):

$$E_H = V_{AC,H} = |E_H| \angle (-60^\circ)$$

2. Transfer to paired L coil  $an$ :

$$E_L = \frac{E_H}{a} = V_{an,L}$$

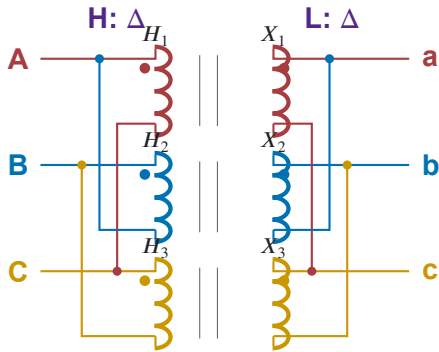
3. Form the L line voltage (Y):

$$V_{ab,L} = \sqrt{3}e^{j30^\circ} E_L$$

$$\delta_{LH} = (-60^\circ + 30^\circ) - 0^\circ = -30^\circ$$

**Phase-shift sign depends on coil polarity, delta joining direction, and endpoint pairing.**

# $\Delta$ – $\Delta$ : physical connection



dots at wire junctions mean connected

## Trace the coil endpoints

H coils:  $AC$ ,  $BA$ ,  $CB$ .

Matching L coils:  $ac$ ,  $ba$ ,  $cb$ .

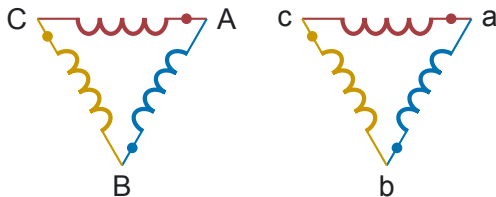
Both deltas use ACB/acb order, with corresponding dots and the same endpoint pairing.

**Both sides use the same cyclic joining.**

# $\Delta$ – $\Delta$ : terminal relations

H:  $\Delta$

L:  $\Delta$



same color = corresponding windings on one magnetic core

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{|E_H|}{|E_L|} = a, \quad \frac{|I_{L,H}|}{|I_{L,L}|} = \frac{1}{a}, \quad \delta_{LH} = 0^\circ$$

**Matching delta connections preserve the paired-coil ratio and angle.**

# Four connections: summary

Delta coil directions, listed by dotted phases  $A, B, C$ :

ABC:  $AB, BC, CA$ ; ACB:  $AC, BA, CB$  (lowercase on L).

| H-L                 | H $\Delta$ order | L $\Delta$ order | $ V_{LL,H} / V_{LL,L} $ | $ I_{L,H} / I_{L,L} $ | $\delta_{LH}$ |
|---------------------|------------------|------------------|-------------------------|-----------------------|---------------|
| Y-Y                 | —                | —                | $a$                     | $1/a$                 | $0^\circ$     |
| Y- $\Delta$         | —                | abc              | $\sqrt{3}a$             | $1/(\sqrt{3}a)$       | $-30^\circ$   |
| Y- $\Delta$         | —                | acb              | $\sqrt{3}a$             | $1/(\sqrt{3}a)$       | $+30^\circ$   |
| $\Delta$ -Y         | ABC              | —                | $a/\sqrt{3}$            | $\sqrt{3}/a$          | $+30^\circ$   |
| $\Delta$ -Y         | ACB              | —                | $a/\sqrt{3}$            | $\sqrt{3}/a$          | $-30^\circ$   |
| $\Delta$ - $\Delta$ | ABC              | abc              | $a$                     | $1/a$                 | $0^\circ$     |
| $\Delta$ - $\Delta$ | ACB              | acb              | $a$                     | $1/a$                 | $0^\circ$     |
| $\Delta$ - $\Delta$ | ABC              | acb              | $a$                     | $1/a$                 | $+60^\circ$   |
| $\Delta$ - $\Delta$ | ACB              | abc              | $a$                     | $1/a$                 | $-60^\circ$   |

Fixed dots; balanced positive supply sequence;  $\delta_{LH} = \angle V_{ab,L} - \angle V_{AB,H}$ .

**Delta joining order changes phase shift; the magnitude ratios remain unchanged.**

# Industry notation: vector groups

## Read Dyn1

**Dyn1**

D: H winding is delta.

y: L winding is wye.

n: L neutral terminal is brought out.

1: L lags H by 30°.

L: 11 H: 12 L: 1



Each clock hour = 30° lag.

## Examples in IEC notation

| Group | H-L             | $\delta_{LH}$ |
|-------|-----------------|---------------|
| Dyn1  | $\Delta-Y$      | $-30^\circ$   |
| Dyn11 | $\Delta-Y$      | $+30^\circ$   |
| Yd1   | $Y-\Delta$      | $-30^\circ$   |
| Dd0   | $\Delta-\Delta$ | $0^\circ$     |

11 means 330° lag, equivalent to 30° lead.

*n* identifies neutral access; specify grounding separately.

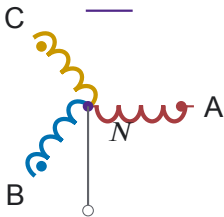
IEC 60076-1: connection symbols.

IEEE C57.12.70: terminal markings and connections.

**Specify winding connections, neutral availability, and phase displacement together.**

# Y: neutral grounding

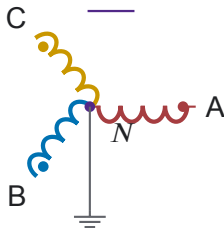
## Ungrounded neutral



The star point  $N$  exists.  
The neutral is intentionally ungrounded.

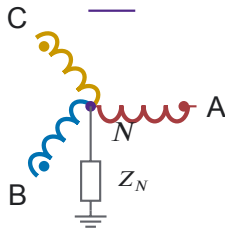
$Z_N$  ( $Z_n$  for a low-side  $n$ ): neutral grounding impedance.

## Solid grounding



$N$  is bonded to ground.  
No added neutral grounding impedance.

## Impedance grounding



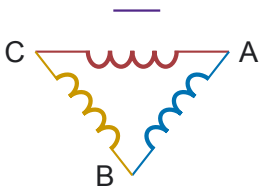
$Z_N$  connects  $N$  to ground.  
The impedance limits ground-fault current.

**Y defines a star point; its neutral branch defines the grounding arrangement.**

Source: Glover et al., sixth edition (2017), Fig. 3.16; Eaton, Power Distribution Systems, Grounding

# $\Delta$ : grounding choices

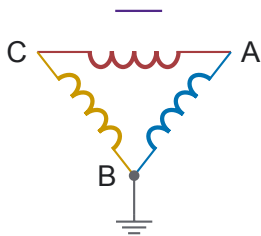
## Ungrounded delta



No star-point neutral.

No intentional ground connection.

## Corner-grounded delta



Phase *B* is bonded to ground.

*B* remains a phase conductor, not a neutral.

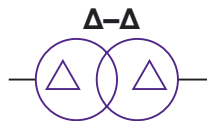
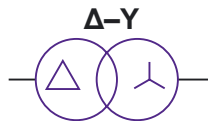
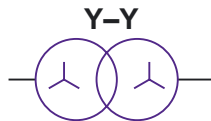
A separate grounding transformer can derive a neutral for a delta system.

**A delta can be grounded, but it has no inherent star-point neutral.**

Source: Schneider Electric, FAQ FA123842; Eaton, Power Distribution Systems, Grounding

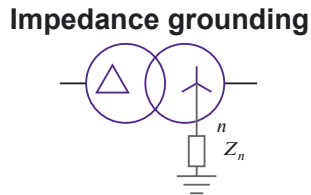
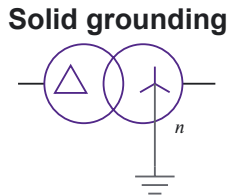
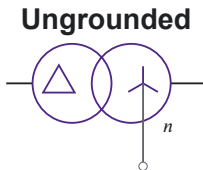
# Single-line symbols and grounding

Connection symbols inside the circles (H left; L right)



Grounding is not specified in this row.

**Grounding branch: the same Δ-Y connection**



**Y/Δ inside the circles; grounding on a separate branch.**

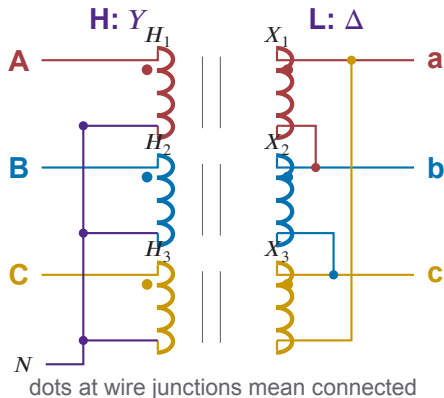
Source: IEEE Std 315-1975, Secs. 6.4.15.1 and 1.9.7.1 (course-redrawn)

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## PART 3

# Example: a three-phase transformer

# Example: given data



**Given: a three-phase transformer**

Rated  $S_{3\phi} = 60$  MVA.

Rated phase-coil voltages:

$$|E_H| = \frac{138}{\sqrt{3}} = 79.6743 \text{ kV}$$

$$|E_L| = \frac{13.8}{\sqrt{3}} = 7.96743 \text{ kV}$$

Ideal, balanced positive sequence.

At rated winding voltages.

Angle reference:  $\angle V_{AB,H} = 0^\circ$ .

**Given coil ratings and connections; find the three-phase terminal quantities.**

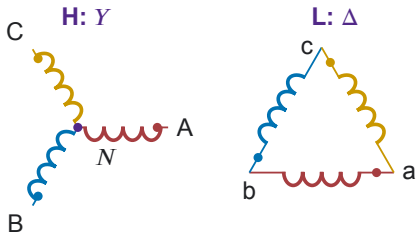
# Example: find and verify

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- a** Find the winding turns ratio  $a$ , the apparent-power rating per phase, both rated line-voltage magnitudes, and their H/L quotient.
- b** Find the low-side line-voltage phasor  $V_{ab,L}$  and the displacement  $\delta_{LH}$ , using  $\angle V_{AB,H} = 0^\circ$ .
- c** Find the rated line-current magnitudes on both sides. Verify  $S_{3\phi} = \sqrt{3}|V_{LL}||I_L|$  independently at H and L.
- d** Reconnect the same coils Y–Y,  $\Delta$ –Y, and  $\Delta$ – $\Delta$  using the Part 2 terminal conventions. Find both line voltages, both line currents, and  $\delta_{LH}$  for each.

**Use coil relations first; check the terminal results with the three-phase rating.**

# Solution (a): ratings and line voltages



Apply one line-to-coil rule per side

$$|V_{LL,H}| = \sqrt{3}|E_H| = 138 \text{ kV}$$

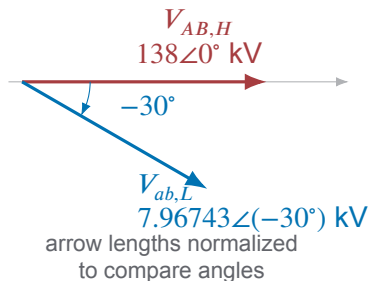
$$|V_{LL,L}| = |E_L| = 7.96743 \text{ kV}$$

$$\frac{|V_{LL,H}|}{|V_{LL,L}|} = \frac{138}{7.96743} = 17.3205$$

The winding turns ratio is still  $a = 10$ .

**The Y-side line-voltage factor turns a coil ratio of 10 into a terminal ratio of 17.3205.**

# Solution (b): voltage phasors



Trace the red coil pair

$$V_{AB,H} = 138\angle 0^\circ \text{ kV}$$

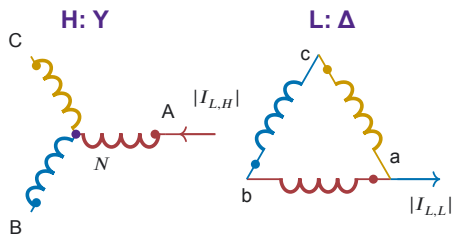
$$V_{AN,H} = \frac{138}{\sqrt{3}}\angle(-30^\circ) \text{ kV}$$

$$V_{ab,L} = \frac{V_{AN,H}}{10}$$

$$\delta_{LH} = -30^\circ - 0^\circ = -30^\circ$$

The transformer preserves the paired coil angle; forming the H line voltage creates the  $30^\circ$  displacement.

# Solution (c): rated currents



$$S_{\phi} = \frac{60}{3} = 20 \text{ MVA}$$

$$|I_{\text{coil}}| = \frac{S_{\phi}}{|E|}$$

## From coil ratings to terminal ratings

|                          | H (Y)   | L ( $\Delta$ ) |
|--------------------------|---------|----------------|
| Coil voltage (kV)        | 79.6743 | 7.96743        |
| Coil current (A)         | 251.022 | 2510.22        |
| Line/coil current factor | 1       | $\sqrt{3}$     |
| Line current (A)         | 251.022 | 4347.83        |
| Line voltage (kV)        | 138     | 7.96743        |
| Total rating (MVA)       | 60      | 60             |

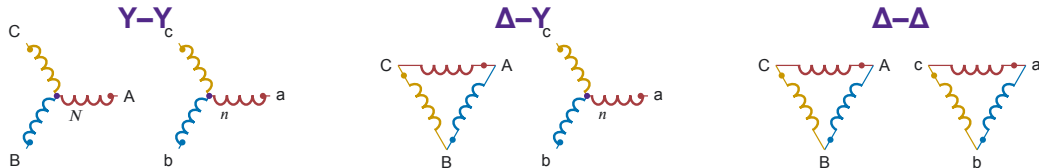
$$S_{3\phi} = \sqrt{3}|V_{LL}||I_L|.$$

Using kV and kA gives MVA on each side.

$17.3205 \times 0.057735 \approx 1$ : the terminal voltage and current factors compensate.

# Solution (d): reconnect the coils

Keep the same coil ratings:  $a = 10$ ,  $S_{3\phi} = 60$  MVA.



| Connection          | $ V_{LL,H} $ (kV) | $ V_{LL,L} $ (kV) | $ I_{L,H} $ (A) | $ I_{L,L} $ (A) | $\delta_{LH}$ |
|---------------------|-------------------|-------------------|-----------------|-----------------|---------------|
| Y-Y                 | 138               | 13.8              | 251.022         | 2510.22         | $0^\circ$     |
| $\Delta$ -Y         | 79.6743           | 13.8              | 434.783         | 2510.22         | $-30^\circ$   |
| $\Delta$ - $\Delta$ | 79.6743           | 7.96743           | 434.783         | 4347.83         | $0^\circ$     |

Same coil ratings; new terminal values; both sides retain 60 MVA.

# Summary

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- 1 Identify the winding sets, phase system, and physical construction.
- 2 Trace matching coil endpoints before reducing the connection diagram.
- 3 Apply Y/ $\Delta$  voltage and current factors: the example has  $a = 10$  but a Y- $\Delta$  terminal voltage ratio of 17.3205.
- 4 Use the terminal phasors to obtain  $\delta_{LH} = -30^\circ$ ; verify the 60 MVA rating on both sides.

**Balanced bank models, autotransformers, and taps use these winding and terminal relations.**